

In this study of the business communication that connects an organization with others in its environment, we link boundary spanning with network theory and propose the concept of an extended network of communication. This extended network is the structure of business communication which flows in work-related ties that organization members maintain both within the organization and across its boundary. We study the relationship between boundary-spanning communication and individual influence in a network with 108 organizational members. We find that boundary spanning correlates with influence, regardless of hierarchical level. There is also a curvilinear relationship between boundary-spanning communication and individual influence. Thus, managers need to balance their communication within and across the organizational boundary.

Keywords: boundary-spanning communication, managerial communication, extended organizational network, influence.

Balancing Ties: Boundary Spanning and Influence in the Organization's Extended Network of Communication

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Contemporary organizations are increasingly characterized by openness to the environment. Permeable organizational boundaries allow and encourage open communication with external entities not only for top

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managers, as in classical bureaucracies, but also for many other managers and employees. For example, organizational members from operations and R&D directly engage in business communication with customers, rather than relying on special boundary-spanning units such as sales or customer service, and contact suppliers in "just-in-time" delivery systems as opposed to maintaining large inventories.

This opening of the organizational boundaries is carried out by managers and employees who communicate on a regular basis with individuals from other organizations—customers, suppliers, government regulators, financing agencies, competitors, strategic allies, universities, consultants. As extensive as research on organizational communication is, scholars have had the tendency to examine communication inside the organization (cf. Nelson & Mathews, 1991; Papa & Papa, 1992; Shah, 1998; Wilson & Malik, 1995) and between organizations (cf. Gulati, 1995; Human & Provan, 1997; Monge et al., 1998) separately. It is somewhat surprising that these two types of communication have rarely been compared or studied together.

In this paper, we consider both external and internal communication. We introduce the concept of an extended organizational network that consists of communication both within and without the organization. Such a network approach to the study of boundary spanning and business communication in general is warranted by the increased importance of communication with external entities for contemporary organizations. We seek to answer three research questions. First, who in the organizational hierarchy engages in boundary-spanning communication? Second, is there a relationship between boundary spanning and individual influence? Lastly, if there is a relationship, what is its form? We argue that our framework is better suited to the study of communication networks of today's organizations than previous approaches.

This paper is organized as follows. We first review the theory on boundary-spanning communication. Then we link the concept of boundary spanning with the theory of social networks and define the organization's extended network of communication. On this basis, we develop hypotheses regarding the relationship of boundary spanning in the extended network and individual influence. We test these hypotheses with network data from an organization, report the results, and discuss their implications for theory and managerial practice.

Boundary Spanning and Contemporary Organizations

The importance of communication across organizational boundaries has increased over the last two decades. Today's information era has brought the advent of the network organization which consists of vertically disaggregated business functions (modules) coordinated by a central unit or broker (Miles & Snow, 1986). These modules would have been

departments or divisions within a single large bureaucracy in earlier days, but are now often structured as separate organizations linked together in a dynamic network. This network organization is coordinated through a specific mechanism distinct from both markets and hierarchies (Powell, 1990). Because these organizations "reengineer" themselves (Hammer & Champy, 1993) to emphasize their own competitive advantage and discard all business processes in which they are not efficient, there is a "virtual effect" produced by the interaction of highly viable separate organizations. That is why the dynamic network they form is sometimes described as a "virtual corporation" (Davidow & Malone, 1992).

Two features of this new organization stand out. First, the dissolution of the hierarchical model of organization has been accompanied by a changing relationship between the organization and the individual (Tsui, Pearce, Porter, & Tripoli, 1997). As layers of middle management become redundant and information technology becomes widely accessible, individual contributors become the organization's most valuable resource (Pfeffer, 1998). Second, contemporary organizations have highly permeable boundaries allowing substantial boundary-crossing communication. So extensive is this communication across the organizational borders that researchers even describe the new model as the "boundaryless organization" (Dess, Rasheed, McLaughlin, & Priem, 1995).

If the importance of boundary-crossing communication is increasing, then communication researchers and manager practitioners, even those in more traditional hierarchical organizations, should pay more attention to it. The concept of boundary spanning (Adams, 1976) suggests that extensive communication is carried out through individual ties crossing the organizational boundary and connecting organization members with members of external organizations. These are primarily communicative linkages that organizational members maintain to "monitor, exchange with, or represent the organization to its environment" (Monge & Eisenberg, 1987, p. 313). Boundary spanning is important to organizations because the flow of information it generates helps them better understand and respond to turbulent environments (Dollinger, 1984). Communication between organizational members and external agents is also a key channel for the organization to influence its sociopolitical environment (Finet, 1993; Seiter, 1995). Some organizations create specialized boundary-spanning departments to handle investor relations and criticisms from outsiders (Rao and Sivakumar, 1999). Boundary spanning is especially important in high technology sectors because the flow of environmental information across the organizational boundary facilitates innovation (Liebeskind, Oliver, Zucker, & Brewer, 1996).

For individual organization members, boundary spanning is important because it provides critical access to decision making (Floyd & Wooldridge, 1997). Research has established that boundary spanners are influential in

organizations because of their ability to scan the environment and process this information (Tushman & Scanlan, 1981a). It results in greater work involvement and increased interaction with superiors, which ultimately leads to reduced turnover (Katz & Tushman, 1983). Boundary spanners become especially powerful when their roles are not routine and inter-organizational transactions are costly and hard to predict (Aldrich & Herker, 1977).

A Social Network Perspective on Boundary Spanning

Boundary spanning is important to members of the organizational network of communication. Social network theory studies the pattern of communication among the members of a given social group as a structure (Lincoln, 1982; Tichy, Tushman & Fombrun, 1979). Communication flows through the contacts that social actors maintain with one another. Network research distinguishes between focal actors ("egos" in network terms, e.g., respondents of surveys) and those with whom focal actors communicate ("alters"). Both types of actors are presented as points or nodes in a network, and communication as links between them. The networks that individual social actors maintain (ego-centered networks) are intertwined in a complex pattern that in its totality represents the whole social network of communication. Communication networks are relatively stable and "can be used to interpret the social behavior of the persons involved" (Mitchell, 1969, p. 2). Studies in this tradition usually focus on a single network under the assumption that the findings can be cautiously generalized to other organizations because they report concepts and examples of established relationships between variables that can be used in analyses of and comparisons with other organizational networks.

Networks of business communication at the organizational level are traditionally classified as a dichotomy: they are either intra- or inter-organizational (Lincoln, 1982). An intra-organizational network consists of the communication patterns throughout (but only within) the entire organization. Studies of business communication networks typically examine this type of network and are conducted at the level of analysis of an individual within a single organizational network. Thus Wilson and Malik (1995) studied how managers' cognitive characteristics shaped the linkages they formed in the intra-organizational network of a large government organization. Shah (1998) focused on the role of an individual's network position in his search for information and choice of social referents in a brokerage firm. Papa and Papa (1992) examined how an individual's position in the intra-organizational network of an insurance company affected the speed of adoption of an innovation. Johanson (2000) researched the mechanisms of social influence in the employee network of a social and health sector organization. While these studies elucidate various important determinants and consequences of communication networks, by design their approach cannot be used to examine boundary spanning.

Alternatively, the study of inter-organizational networks is often at the level of analysis of whole organizations. Such networks represent an important infrastructure through which organizations produce and distribute information to facilitate collective action in alliances (Monge et al., 1998). The structure of an inter-organizational network can improve the effectiveness of its participant organizations by facilitating interactions among them (Human & Provan, 1997). Communication in inter-organizational networks breeds trust, which influences the type of alliances that organizations form (Gulati, 1995). Inter-organizational research seeks to enhance our understanding of communication between organizations, but it does so, generally, by excluding communication within these organizations.

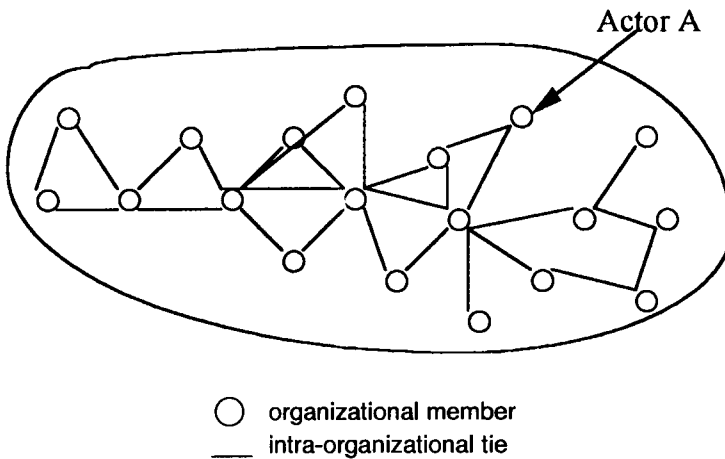
To study boundary spanning, it is necessary to bridge the two types of networks, intra- and inter-organizational. Previously, communication between individuals belonging to different organizations has been examined as overlapping memberships on boards of directors (Gulati & Westphal, 1999; Koenig & Gogel, 1981) and as connections among people in labor markets that extend across organizations (Althauser, 1989). Knowledge of boundary spanning can be facilitated by research in this vein if it can compare properties of both intra- and inter-organizational networks.

Given the increased importance of individual organization members in contemporary organizations, we focus on the individual level of analysis in our search for an appropriate way to study boundary-spanning communication and the extended network it sustains. In contrast to prior studies, we consider the business communication of managers with recipients both inside and outside the organization. Some of the external work ties through which business communication flows can be quite stable and lasting, and they are often critical for the organization, such as relationships with a major corporate supplier or a government agency. The boundary-spanning communication between individuals is embedded (Granovetter, 1985) in the larger relationship between their respective organizations. Communication among individuals from different organizations has been shown to have an effect on the ties between those organizations (Johnson et al., 1996). Extending this line of inquiry, we focus on the relationships between boundary spanning and both hierarchical rank and the influence of managers in an organization.

Defining an Extended Network of Organizational Communication

Boundary specification is a major problem which network theory has yet to resolve conclusively (Monge & Eisenberg, 1987). Networks are delineated according to two alternative perspectives: realist and nominalist (Laumann, Marsden, & Prensky, 1983). According to the realist position, the network's borders are defined where the research subjects themselves perceive them. In the nominalist view, the researcher defines where the

Figure 1
Internal Network

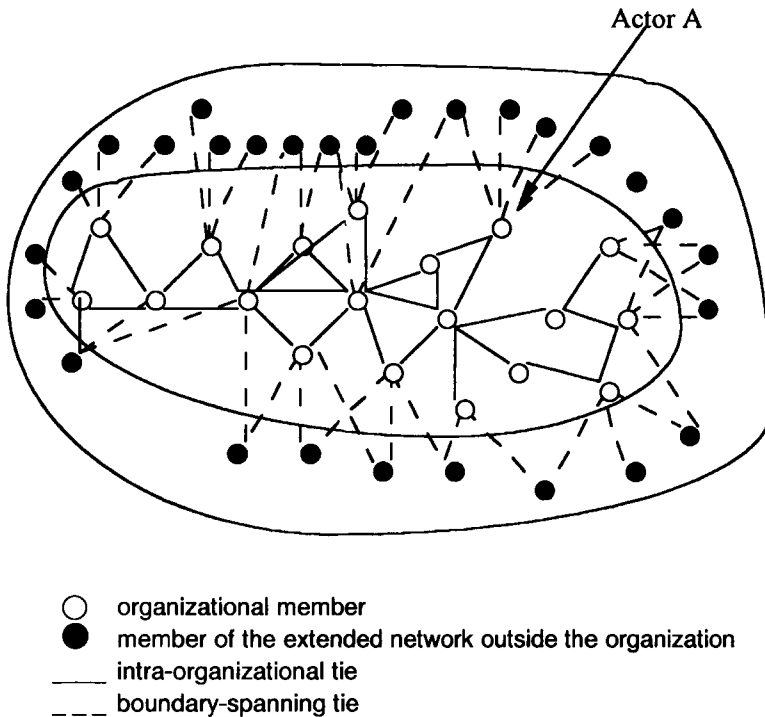


network ends based on the conceptual framework in use. In almost all research on organizational networks, the two approaches converge on selecting the organization's boundary as the boundary of the network. The traditional intra-organizational network that has been used (see Figure 1) does not allow the study of boundary spanning and therefore entirely excludes boundary-spanning communication with all its volume, extent, and importance.

Since contemporary organizations are characterized by increasingly permeable boundaries and have a relatively high share of boundary-spanning communication, the study of business communication needs to account for these boundary-crossing contacts. That is why we propose the concept of an extended network, in which relatively stable boundary-crossing ties are included in addition to the internal network (see Figure 2). The inclusion of boundary-spanning ties in the organizational network can be explained using either the realist or the nominalist perspective on boundary delineation. In the eyes of the organization's members (realist view), relatively lasting ties with outsiders can be at least as important for the workflow as ties with insiders. For the researcher (nominalist view), the study of boundary-spanning communications becomes appropriate in light of recent organizational theory regarding the opening of contemporary organizations toward their environment.

The extended network of communication thus consists of intra-organizational and inter-organizational ties examined at the individual level. Taking its cue from the term "boundary spanning," this conceptualization

Figure 2
Extended Network



of the organizational network literally spans its boundaries, extending them to include those individuals who are outside the organization but important for its activity and with whom organization members interact on a regular basis. The extended organizational network has the internal network as its core. It resembles an inter-organizational network inasmuch as it includes communication external to the focal organization. At the same time, similar to an intra-organizational network, it focuses on ties at the individual level. The extended network is also akin to an ego-centered network, but with an organization being the ego.

There are two types of actors in the extended network: primary actors within the organization and secondary actors who are outsiders connected with the organization. Outsiders may or may not be connected among themselves. Organization members often have different characteristics in the two networks. Consider Actor A who is relatively marginal in the internal network (see Figure 1) with only two contacts with other individuals. However, this actor interacts regularly with five more individuals

outside the organization (see Figure 2). Actor A's "degree centrality" (Freeman, 1979), measured as the number of connections to others in the internal network, is 2, but it is 7 in the extended network. The extended network then renders a better description of Actor A's business communication. Further, we introduce the term *boundary-spanning orientation*, or the proportion of boundary-spanning ties in a manager's total number of ties in the organization's extended network, as a measure of external contacts. Actor A's boundary-spanning orientation is 5 external contacts divided by 7, or .71, which is obviously fairly external. Examining managers' characteristics in the extended organizational network can help answer the question of whether boundary spanning creates influence within the organization for individuals in such roles.

Next, we develop hypotheses regarding communication networks and influence. Individual influence in organizations is obviously strongly related to hierarchical level. To be able to concentrate on the effect of boundary-spanning communication on influence, we first describe the hypothesized effect of hierarchical level on boundary spanning. Next, we propose that position in the extended network is related to influence, beyond the effect of hierarchical level. Finally, we hypothesize about the form of the relationship between boundary-spanning ties and influence.

Hierarchical Level and the Extended Network

Where one is placed in the organizational hierarchy is likely to affect opportunities for establishing internal and external organizational ties. Classic studies of management have established that activities of managers vary with hierarchical level, with top management typically determining strategy and providing legitimacy through extensive interactions with the environment. CEOs spend a large share of their time in the externally oriented functions of figurehead, spokesperson, and monitor (Mintzberg, 1973). Top managers have a responsibility for the formulation of strategy, which is heavily influenced by their boundary-spanning ties (Geletkanycz & Hambrick, 1997). External ties enhance the career building potential for managers (Krackhardt, 1990). The greater number of external contacts that top managers maintain in comparison with other managers is reinforced through their membership in different external clubs and societies (Carroll & Teo, 1996).

Since middle managers maintain ties with both the top and lower-level managers (Stevenson, 1990), there may be a "division of responsibility" among the upper and middle levels of management. Top management may focus mostly on networks across the organizational boundary, while middle management maintains more communication within the organization—both with top managers and with lower managers. Further down the hierarchy, lower managers who implement strategic decisions made at upper levels may concentrate mostly on internal networks.

Even if contemporary organizations are considered flatter than traditional bureaucracies, managers higher in the hierarchy—to the extent that a hierarchy remains—would be likely to maintain a larger absolute number of external ties than lower-level managers. We expect that boundary-spanning communication will be stronger for upper management also in relative terms, that is, compared to the number of their internal contacts. Even if managers at higher levels may be more centrally located in the internal network, external contacts should comprise a larger share of their ego-centered networks. Therefore, we offer the following hypotheses:

- H1: Managers at higher levels of the hierarchy have a higher number of external contacts than managers at lower levels.
- H2: Managers at higher levels of the hierarchy have a more external boundary-spanning orientation than managers at lower levels.

Boundary Spanning and Influence

Research suggests that there may be a positive relationship between boundary spanning and individual influence (Schwab, Ungson, & Brown, 1985). Managers in boundary-spanning subunits have higher influence because of their ability to mediate between the external environment and internal decision making (Floyd & Wooldridge, 1997). This is especially true for the high technology industries where the speed of obtaining information is critical for organizational flexibility (Liebeskind et al., 1996). However, boundary spanning requires that an individual direct his or her efforts outside the organization, away from the control of internal resources. Thus, some studies find no evidence of a significant relationship with influence (cf. At-Twajri & Montanari, 1987).

According to a resource dependency perspective (Pfeffer & Salancik, 1978), individuals strive to maximize their control over resources which results in higher influence. One way to achieve such control over resources is through the organization's social network. Research has established that network centrality is related to influence. Actors with high centrality have easy access to and potential control of informational and physical resources (Brass, 1992; Fombrun, 1983). This centrality is positively associated with influence. Numerous studies have documented this relationship for the intra-organizational, or internal, network (cf. Brass, 1984; Burkhardt & Brass, 1990; Krackhardt, 1990), but if outside interactions are an important channel for obtaining environmental information and gaining stature, then organization members are likely to seek boundary-spanning opportunities. Boundary spanning need not be an activity performed only by those higher up in the hierarchy. On the contrary, individual organizational members can use boundary spanning as a resource to enhance their careers. That is why centrality in the extended network is expected to be positively associated with influence, and this

effect should be beyond the contribution of hierarchical level. Thus, our next hypothesis:

H3: Centrality in the extended network is positively associated with influence, regardless of hierarchical level.

As previously mentioned, actors who occupy central positions within the organization are influential because of their access to and control over resources. However, the inclusion of ties with outsiders in the extended organizational network makes boundary-spanning personnel more central than they are in the internal network. If organizational boundaries become less clear cut and information flows more freely in contemporary organizations, the extended network should be more relevant for understanding business communication than the traditional intra-organizational network. To the extent that a manager's boundary-spanning communication is an important component of his/her work, a manager's position in the extended network should be more indicative of this individual's influence than position in the internal network. Thus our next hypothesis:

H4: Centrality in the extended network is more strongly and positively associated with influence than centrality in the organization's internal network.

Balanced Networks and Influence

The transmission of information from the environment throughout the organization is a two-step process in which information is first acquired from outside sources and then passed on to organization members (Tushman & Scanlan, 1981b). The amount of information transferred is thus a function of the number of both external and internal ties. Tushman and Romanelli (1983) established that those individuals connected both inside and outside the organization were the most influential, but those with ties only to the outside were the least influential in their sample. Obviously, organization members who are connected mostly internally would not be effective boundary spanners. By the same token, an individual who is connected mostly to the outside, with only one or two contacts within the organization, would have such an externally oriented network that he/she would not have much opportunity to transfer environmental information within. Even if such an individual collected massive information from the environment, lack of ties within the organization would not allow him/her to be an effective boundary spanner and therefore reap the benefits of increased influence. It seems logical to expect, then, that effective boundary spanning would be done by people who are well connected on both the inside and the outside of the organization, and that individuals with relatively equal numbers of internal and external ties would be most likely to facilitate an efficient transfer of environmental information and accrue influence. Thus, our final hypothesis:

H5: Individuals with balanced networks are more influential than those with either internally or externally oriented networks.

Methods

These hypotheses were tested using data gathered from 108 managers in an urban transit authority in the western United States. All managers were surveyed, and the response rate was 79%. The survey asked questions about background characteristics, followed by a sociometric instrument that instructed the respondent to list individuals with whom he/she communicated "to get the job done" both within and outside the organization. To aid in recall, respondents were provided with a list of all other managers in the network. Data about the frequency of these interactions were also collected. The questionnaire is reproduced in part in the Appendix.

The organization studied here, as is typical for transit authorities in the US, is a publicly held organization with a multi-layered hierarchy, a functional structure, and relatively complex coordination mechanisms. At the same time, managers engage in boundary-spanning communication outside the organization, allowing an analysis of the relationship between formal position, boundary spanning, and influence. Recent research has used similar organizations in the public sector as appropriate sites to study both business communication (Suchan, 1998) and social networks (Johanson, 2000; Toonen, 1998).

Independent Variables

Hierarchical *level* reflects the three existing major levels of management in the organization and is measured as top, middle, and lower management. Centrality, which is the most important network indicator of structural location at the individual level, is measured by outdegree (Freeman, 1979). *Outdegree* stands for the number of direct contacts that an individual initiates and therefore measures potential communication activity. It is measured in both the internal and extended networks of the sample organization. When analyzing the extended network, we assumed, for the purposes of this study, that outsiders are not connected with other outsiders.¹

As earlier defined, the *orientation* of an individual's network is the number of boundary-spanning ties divided by the total number of ties (external plus internal). A high value (up to 1) stands for external orientation, whereas a zero score indicates an entirely internal orientation.

Dependent Variables

Individual influence is measured by two variables. The first one, suggested by Van de Ven and Ferry (1980), measures *self-perceived influence* in decision making. It is computed as the sum of the responses (on a 5-point scale) to questions about the influence the respondent has over

Table 1
Descriptive Statistics and Correlations (N = 108)

Variable	Mean	SD	Min	Max	1	2	3	4	5	6
1. Hierarchical level	1.60	.76	1	3						
2. Outdegree in internal network	11.31	3.60	1	14	.26					
3. Boundary spanning ties	3.60	3.32	0	9	.32	.18				
4. Outdegree in extended network	14.91	5.76	1	23	.35	.84	.82			
5. Boundary spanning orientation	.20	.13	0	.67	.15	-.03	.78	.46		
6. Self-perceived influence	3.05	1.25	1	5	.60	.35	.34	.42	.14	
7. Attributed influence	16.51	13.99	0	63	.67	.32	.32	.39	.15	.65

Note. Correlations above .20 are significant at .05 level.

determining the goals for the work unit, policies and procedures, and decisions about hiring, promotion, and salary increases (Cronbach's alpha = .91). Since self-perceived indicators may be biased, a second measure of influence is added—*attributed influence*. It measures the influence that other organization members attribute to the respondent. To calculate it, we asked respondents, or egos in network terms, to rate how much influence every manager with whom they communicated (alter) had on their work performance. These ratings were then summed for every alter, and these sums measure attributed influence. This is a single-item measure which is significantly and strongly correlated with the first measure for the dependent variable, self-perceived influence ($r = .65, p < .001$).

Results

The respondents in our sample were 59% male and 41% female, 78% Caucasian, with an average age of 38 years. Descriptive statistics and correlations among the major variables are shown in Table 1. The managers in the sample maintain regular boundary-spanning contacts with a total of 324 people outside their organization, and in some cases two or more managers have ties with the same outsider. They have an average of 11.3 contacts with other organization members (outdegree), and 3.6 boundary-spanning ties. In other words, about one of every four ties that individuals maintain is outside the organization. Most boundary-spanning ties occur along the lines of functional responsibilities: for instance, managers

Table 2
Formal Status and Boundary Spanning:
ANOVA and Multiple Comparisons

Independent Variables	Hierarchical Level		
	Lower Management (N = 61)	Middle Management (N = 29)	Upper Management (N = 18)
Number of boundary spanning contacts (S.E.)	2.76 (.41)	4.10 (.56)	5.67 (.75)
$F = 6.27, p < .003$	$p < .065$		$p < .107$
$Eta = .32$	$p < .004$		
Boundary spanning orientation (S.E.)	.17 (.02)	.22 (.03)	.28 (.03)
$F = 3.16, p < .046$	$p < .211$		$p < .220$
$Eta = .24$	$p < .016$		

from the Governmental Relations department communicate with people from the state, county, and city regulatory agencies and the transit agencies of other cities; those from Transit Services communicate with representatives of auto parts stores, garages, fuel providers, and maintenance services; managers from Finance and Administration maintain communication links with accountants and outside marketing specialists. Only two communication ties were reported with researchers from universities, and six with consultants. Most of the ties in the internal network are relatively strong: 63% of respondents communicated at least once or twice a week with their alters concerning work. In contrast, only 39% of boundary-spanning ties were that strong.² Such predominantly internal orientation is consistent with the general image of public bureaucracies as organizations that are relatively mechanistic and inwardly focused.

Hypothesis 1 states that managers at higher levels have more work-related contacts across the organizational boundary than those at lower levels. Variance was tested for homogeneity,³ and no significant differences between groups were found. Therefore a one-way ANOVA was appropriate for the test of Hypothesis 1 (see Table 2). It showed a significant difference in the number of boundary-spanning contacts among the three management tiers ($F = 6.27, p < .003$). Mean differences across groups were tested using Scheffé's post-hoc method for multiple comparisons, which requires a higher threshold of difference than most other methods in order for results to be considered significant. Mean numbers for the three groups were arranged in the expected direction (highest for top

Table 3
Effect of Boundary Spanning on Influence (N = 108)

Independent Variables	Self-perceived Influence (betas)			Attributed Influence (betas)		
	(1)	(2)	(3)	(4)	(5)	(6)
Hierarchical level	.60***	.55***	.51***	.67***	.63***	.61***
Outdegree (internal network)		.21**			.15*	
Outdegree (extended network)			.24**			.18*
R^2	.36**	.40***	.41***	.45***	.46*	.48*
$F(\Delta R^2)$		7.01**	8.97**		4.44*	5.58*

*** $p < .001$

** $p < .01$

* $p < .05$

managers and lowest for low-level managers). The test indicated a significant difference ($p < .004$) between the first and third group, while the other comparisons only approached significance. Therefore there is qualified support for Hypothesis 1.

Table 2 also reports the results of the test of Hypothesis 2 according to which managers at higher levels of the hierarchy would have a more external orientation than managers at lower levels. Similar to the results for Hypothesis 1, variance was found to be homogenous across the three groups of the independent variable. A one-way ANOVA indicated a significant difference in the number of boundary-spanning contacts among the three management tiers ($F = 3.165$, $p < .046$). Scheffé's test was significant ($p < .016$) again only for the comparison between the first and the third groups of managers, lending partial support for Hypothesis 2. Thus managers higher up in the hierarchy appear to engage in more boundary-spanning contacts not only in absolute terms but also relative to the number of their internal ties, even if they are generally more centrally located in the internal network.

Table 3 presents the tests of Hypotheses 3 and 4, which were performed using regression analysis. Hypothesis 3 states that centrality in the extended network will be positively associated with influence, regardless of hierarchical level. The base models 1 and 4 indicate that self-perceived and attributed influence rise with hierarchical level. Models 3 and 6 demonstrate that centrality in the extended network is positively and significantly associated with self-perceived and attributed influence (*betas* of .24, $p < .01$ and .18, $p < .05$, respectively), controlling for hierarchical level. As demonstrated by the F test of the increase in R^2 , this effect is

beyond the contribution of hierarchical level in the overall model. Hypothesis 3 is therefore supported.

However, the test for Hypothesis 4, which predicted that centrality in the extended network would be more strongly associated with influence than centrality in the organization's internal network, was inconclusive. As shown in Table 1, the two indicators for centrality in the internal and broader network are strongly correlated ($r = .84, p < .001$). To avoid multicollinearity, they are entered in separate regressions in Table 3. Extended outdegree explains slightly more variance in self-perceived and attributed influence (R^2 of .41 versus .40 for self-perceived influence in models 3 and 2, and an R^2 of .48 versus .46 in models 6 and 5), although the difference is relatively small. The first-order correlation coefficients of extended outdegree compared to internal outdegree (Table 1) with influence are likewise slightly larger (.39 versus .32, and .42 versus .35). However, a test of the difference between the correlation coefficients (Montgomery & Peck, 1982) fails to reject the null hypothesis.

Finally, Hypothesis 5 states that individuals with balanced networks are more influential than those with internally or externally oriented networks. The hypothesized relationship is a curvilinear inverted U-shaped function described by a quadratic equation:

$$y = b_0 + b_1x + b_2x^2,$$

where:

y denotes predicted influence,

x is orientation of an individual network, and

$b_2 < 0$.

This is tested through a hierarchical polynomial regression procedure (Table 4). A linear regression ($y = b_0 + b_1x$) is tested in the first step, and the quadratic term x^2 is introduced in the second. The partial regression coefficients are significant in the expected direction for self-perceived influence, but the coefficient for the quadratic term b_2 for attributed influence is not significant. The F -test of the increase in R^2 for self-perceived influence indicates that the addition of the quadratic term results in a significant improvement of the fit of the model over the simple linear regression. The fitted line is $y = 2.5 + 6x - 9.7x^2$, which peaks at a value of .61 for orientation. The relationship between boundary-spanning orientation and influence is then curvilinear—influence rises until a moderately high value of orientation, and slightly decreases for extremely high values. These results present partial support for Hypothesis 5 that balanced boundary-spanning orientation is related to highest influence.

Discussion

In this study, we defined an extended organizational network which

Table 4
Linear and Non-linear Effects of Boundary Spanning Orientation on
Influence ($N = 108$)

Independent Variables	Self-perceived Influence (betas)	Attributed Influence (betas)
Boundary spanning orientation	.23*	.18*
R^2	.05*	.05*
Boundary spanning orientation (Boundary spanning orientation) ²	.80** -.60*	.57* -.38
R^2	.10**	.06
$F(\Delta R^2)$	5.08*	2.00

** $p < .01$

* $p < .05$

includes boundary spanning in addition to communication within the organization itself. This approach contributes to a better understanding of the relation between boundary spanning and individual influence. Specifically, our results provide at least partial answers to three research questions: (a) who in the organizational hierarchy is involved in boundary spanning; (b) whether boundary spanning is related to influence beyond the effect of one's hierarchical level; and if so, (c) what the form of this relationship is.

This study confirms that, even in a traditional bureaucracy, most managers engage in business communication across the organizational boundary. As expected, managers at higher hierarchical levels engage in a larger volume of such external networking, both in absolute terms and relative to their internal contacts. There are at least two alternative explanations for this finding. First, if individuals strive to maximize their control over resources and thus their influence (Pfeffer & Salancik, 1978), outside interactions are an important channel for obtaining important environmental information and gaining stature. Therefore, boundary spanning may have helped these upper-level managers to rise to their positions (Krackhardt, 1990). Second, the nature of top managers' work may require that they interact with outside entities more often (Carroll & Teo, 1996).

Generally, it can be argued that the upper levels of management are part of a social elite who prefer to interact with others in their social strata in other organizations. While such an elite social-circle effect might arise in some other social networks, this scenario would probably not apply to the network studied here. This was a relatively small public transit agency. In the United States, it is not unusual for a bus driver to become the head of a transit agency, and this was the case in this agency. All of the respondents studied were managers, and top managers had often been promoted from within departments. Hence status differentials

between levels of management are relatively small in the organization studied here.

Further, we find that an individual's boundary-spanning communication is related strongly and positively with influence, and that is beyond the effect of hierarchical level. As others have found (Floyd & Wooldridge, 1997; Schwab, Ungson, & Brown, 1985), those who are more active in the extended network tend to be more influential. Importantly, centrality in the extended network is at least as good a predictor of influence as centrality in the internal network. However, the extended network that we define in this paper accounts for interactions both within and across the organizational boundary. Therefore the extended network appears to provide a richer and more thorough description of managers' locus of business communication and its relationship with influence.

Finally, the present study suggests a curvilinear relationship between the orientation of an individual ego-centered network (as defined by the share of boundary-spanning ties in the total number of ties that a manager maintains in the organization's extended network) and influence. Those who balance their boundary-spanning and internal contacts appear to be most influential. As Tushman and colleagues (Tushman & Romanelli, 1983; Tushman & Scanlan, 1981b) have suggested, effective boundary spanning requires extensive contacts with individuals both across and within the organization's boundary.

These findings about boundary spanning become more relevant considering the current theoretical developments about contemporary organizations such as network organizations (Miles & Snow, 1986), virtual corporations (Davidow & Malone, 1992), and boundaryless organizations (Dess et al., 1995). Today's organizations are often not single firms but groupings of strategically linked firms (Freeman & Barley, 1990) or "organization sets" in which organizations work closely together with their suppliers, buyers, important regulators, and even competitors (Evan, 1993). All these complex linkages and alliances are carried out through communication between individual members in organizations who perform critical boundary-spanning roles. Thus boundary-spanning communication at the individual level represents the main channel through which inter-organizational networks function, and our research suggests that balancing internal and external contacts is important for individual effectiveness in contemporary organizations.

Implications for Future Research

In this study, we find that, even in a public bureaucracy, an extended organizational network of communication that incorporates individual boundary-spanning contacts describes managers' locus of activity better than the traditional internal network. It is therefore a viable research tool that can be used in further studies with other types of organizations. By

outlining this network and its implications, this study has made a contribution to understanding network boundaries, a recurrent problem in network research (Monge & Eisenberg, 1987).

Further research on the extended network can shed light on several intriguing questions: How is critical environmental information transmitted throughout the organization? How are key technological and managerial innovations adopted? How are alliances formed through boundary-spanning ties? Especially interesting is the question about the role that an individual boundary spanner plays in these processes: For which boundary-spanning roles is a single individual enough and for which roles does the organization need to assign a whole boundary-spanning group? How extensive is communication across the organizational boundaries relative to communication within them, in different functions, organizations, and industries? While we have used a network approach to study the structure of communication, future qualitative studies can examine the messages that are communicated across the organizational boundary and compare them with business communication within the organization.

There were several limitations on this study's method, correlational design, and sample. As is typical for network studies of communication, this study has relied on detailed quantitative data from a single organization. While it has focused on boundary spanning, a necessary activity for any organization, its results should be interpreted with caution. The findings of this research are correlational in nature, and while they generally support the hypothesized relationships, they do not establish causality.⁴ The organization studied here is a publicly held transit authority. While it sought to adopt innovative approaches in its management, it is hardly typical of the contemporary network organizations that are most often described in the high technology (e.g., the software industry, biotechnology) and service (consulting, financial services) sectors. Further research is needed to study boundary-spanning communication in other industries, which undoubtedly will produce many interesting insights into the role and share of boundary-spanning communication. Still, the fact that the organization studied here is a transit authority only strengthens the case for the theoretical arguments we make: If boundary spanning is found to comprise a substantial share of the work-related communication, and the relationship with individual influence is significant in such an organization, then it must be even stronger in information-intensive industries.

Implications for Management

The results presented here imply that managers should think of their organizations in broader terms. The organization is no longer circumscribed by the traditional boundaries but spans them to include important outsiders. To be influential and effective, managers ought not to limit their business communication to those within the organization but should

stretch across its borders—monitor the environment, exchange resources with external organizations, or represent the organization.

Another important practical implication of these findings is that managers need to be effective boundary spanners if they want to increase their individual influence and enhance their careers. Boundary spanners are influential because they are able to scan the environment for critical changes and then communicate this knowledge with superiors, peers, and subordinates. Those who network only internally miss the influence that accrues to boundary spanners. However, managers who network mostly externally are not necessarily influential either. Managers become influential when they are effective boundary spanners who balance communication within their organization with communication with representatives of other organizations.

NOTES

1. This assumption may not be true in all cases—that is, some outsiders may communicate among themselves. Salespeople from different organizations, for example, form selling alliances (Smith & Barclay, 1997) and teams (Moon & Armstrong, 1994) in which they communicate extensively. However, the focus of this study is the relationship between boundary-spanning communication and individual influence *within* an organization. On a *prima facie* basis, communication among outsiders would hardly have a meaningful effect on the influence of managers in the studied organization. The very high cost of collecting data on these ties among outsiders therefore outweighs substantially the possible new knowledge which they can generate.

2. No significant differences between the roles that strong and weak ties play for individual influence were found.

3. The Cochran C and Bartlett-Box F tests, which are standard in SPSS, were used.

4. It is indeed possible that managers with greater influence, perhaps due to charisma, may be selected more frequently for boundary-spanning roles to represent the organization externally. We are grateful to an anonymous reviewer for pointing out this alternative interpretation. In this study, we have adhered to the more conventional view that individual influence and higher positions in the hierarchy are endogenous to an individual's actions at work, including boundary-spanning communication.

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Appendix

Survey Questions

1. Self-perceived influence: How much influence do you have in making each of the following decisions:
 - a. Deciding work goals for your unit?
 - b. Adopting new policies?
 - c. Hiring new employees?
 - d. Promoting employees?
 - e. Salary increases of other employees?

[1 none, 2 little, 3 some, 4 quite a bit, 5 very much]

2. **Internal communication network:** Please list the people at work that you interact with to get the job done and respond to the questions about each one of them.
3. **Boundary spanning communication:** Do you interact with people in other organizations (such as vendors, contractors, government agencies, other transit agencies, etc.) to get your job done?
[If yes] Please print the names and organizations of these people and answer the questions for each one of them.
4. **Frequency of tie:** How often do you communicate with [alter]?
[1 at least once a day, 2 once or twice a week, 3 about every two weeks, 4 about monthly, 5 once or twice in the last six months]
5. **Attributed influence:** How much influence does [alter] have on the performance of your work?
[1 none at all, 2 a slight amount, 3 some, 4 a considerable amount, 5 a great deal]